```
  } else {
    Serial.println("Error opening data.txt.");
  }
}

void loop() {}
```

This sketch builds on concepts elsewhere in this book, including the delay without using the `delay()` function and the Analog Serial Monitor, however, the SD library's functions are a little different from the other libraries that we have looked at.

File

The first line we will look at creates an instance of the file that we want to use. It has a very simple syntax, as follows:

```
File name
```

Where `name` is anything you want it to be. This does not have to be the actual file name on the SD card. In our example code, we used the line `File file;` because it seemed easiest although `myFile`, `dataFile`, or any other name that you like could be used here.

SD.begin()

With our file ready to go, we need to initialize the SD library using the `SD.begin()` function. It too has a basic syntax:

```
SD.begin(csPin)
```

This function will default to a CS, or chip select pin of pin 10, so in order to use this function with the Ethernet Shield or any other configuration where the CS pin on the SD card is on any other pin we need to specify that pin number. From our code it is as follows:

```
if (!SD.begin(sdCsPin)) {
```

This line contains the `SD.begin()` function and performs a test to see if the function was successful or not. If the SD card is successfully started, this function will return `true` while if there was a problem, for example the SD card is not present, it will return `false`. The `if` statement in our code will give us an error message and exit the sketch if there is any problem initializing the SD card.

■ **Note** Even if we are using a different CS pin from the default pin 10, we need to configure pin 10 as an `OUTPUT` or else the SD library will not properly function.

SD.open()

To be able to write data to a file, we must open it first with the `SD.open()` function. The syntax for the function follows: